Code No: 157AA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
AD-HOC AND SENSOR NETWORKS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) State two design challenges of a Manet. [2]
b) List the important characteristics of a Manet. [3]
c) State the role of jitter in any broadcast protocol for Manet. [2]
d) Define and illustrate flooding mechanism. [3]
e) State the applications of geo-cast routing. [2]
f) State the reasons for poor performance of TCP over ad hoc links. [3]
g) Write the formula in LEACH protocol for CH selection. [2]
h) Define-Sensor, Sensor Node and Sensor Network. [3]
i) What is meant by data aggregation in sensor applications? [2]
j) Write about COUGAR, sensor database system. [3]

PART – B

(50 Marks)

2.a) Write about any one location service algorithm for position-based routing in Manets.
b) Illustrate any one forwarding strategy for position-based routing in Manets. [5+5]

OR

3.a) State the merits and demerits of proactive and reactive routing protocols for Manets.
b) Illustrate the working of ZRP, Zone-based Routing Protocol. [5+5]

4.a) What is meant by Broadcast Storm Problem in Manet routing? Illustrate with an example.
b) Differentiate the tree-based and mesh-based multicast schemes for Manets. [5+5]

OR

5.a) Explain the tree-based multicast scheme for Manets, AMRIS.
b) Write the algorithmic steps for MPR (Multi-Point Relaying) protocol for Manet broadcasting. [5+5]

6.a) Giving a neat sketch, write about Mesh-based Geocast Routing (MGR) protocol.
b) Explain the ELFN protocol for TCP over ad hoc links. [5+5]

OR

7.a) What is meant by anycast? Explain the GeoTORA protocol making use of anycast.
b) Explain how COPAS addresses the unfairness issue of TCP in ad hoc environment.[5+5]

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8.a) Giving a architectural diagram, write about Great Duck Island application of wireless sensor networks.

b) Explain how data dissemination in sensor networks is achieved using Directed Diffusion algorithm. [5+5]

OR

9.a) Explain the differences between TEEN and APTEEN protocols in WSNs.

b) Explain the distinction of PEGASIS protocol over LEACH. [5+5]

10.a) Explain how dynamic nature of WSN is addressed with adaptation concept?

b) Write about query processing in wireless sensor networks. [5+5]

OR

11.a) With the help of an architectural diagram, write about any one sensor database system.

b) Write about in-network processing in sensor applications emphasizing its merits. [5+5]

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